

Tidy

K Arnold

Homework 3

- GitHub repos created for you!
- Comment out 2012 data if it's giving you trouble

Facial Recognition (Discussion 2)

**Sit near your team
(see attendance sheet)**

Why cohorts?

- Relationships in a socially distant time
- Teamwork as a "soft skill"
- Learn collaborative workflow using GitHub
- Help each other within a cohort
- Course staff can meet with cohorts together
- Possible final project teams

For each assignment, each *cohort* randomly split into 2 *teams*.

Slow it down...

- HW 3 due date extension
- HW 4 small

Friday

- Reminder: Quiz (including feedback)
- Discussion: reply instructions posted
- HW4 and Prep 4 posted this afternoon.
- Which two cohorts want to meet with me next week?

Facial Recognition Surveillance

- What good points did others raise?

Ethical Frameworks

- Utilitarian ("do the benefits outweigh the harms? to whom?")
- Virtue ("does this align with my core values?")
- Analogical ("is there a simpler situation I can compare this to?")
- Deontological ("is this decision lawful?")

Which did you use? Which did your peers use?

Tidying and Joining Data

Outline:

- the dataset
- tidying
- joining
- plotting

Data wrangling often takes a *lot* of time and effort, so buckle in.

JHU COVID-19 data

As you might imagine, keeping a comprehensive list of all COVID-19 cases worldwide involves pulling data from numerous sources (and frequent updating). Fortunately, some folks at Johns Hopkins have been doing that work and putting the resulting data into a github repository that anyone can access.

Navigating the GitHub Repo

You can visit their github project at

<https://github.com/CSSEGISandData/COVID-19>. There you will find

- details about what sources were used for the data,
- what sorts of data are available, and
- some places the data have been used

You will also see this note:

The Website relies upon publicly available data from multiple sources, that do not always agree.

Finding some data

You could clone the repository, but you also just pull the data directly from their repository. They've split the data into "daily reports" (one CSV per day, all measures) and "time series" (one CSV per measure, all days). Here is an example page showing **one of the data sets** available to you.

CSSEGISandData / COVID-19

Watch

438

Star

9.4k

Fork

3.5k

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Branch: master

COVID-19 / csse_covid_19_data / csse_covid_19_time_series / time_series_19-covid-Confirmed.csv

Find file

Copy path

CSSEGISandData update

5fdea6a11 minutes ago

1 contributor

453 lines (453 sloc)67.2 KB

Raw

Blame

History

Search this file...

1	Province/State	Country/Region	Lat	Long	1/22/20	1/23/20	1/24/20	1/25/20	1/26/20	1/27/20
---	----------------	----------------	-----	------	---------	---------	---------	---------	---------	---------

We want raw data

GitHub renders CSVs in a fancy way, but you can get the plain old CSV if you click the Raw button. We're mostly interested in the URL for this file, since that will let us pull the data into R.

```
confirmed_global_url <- paste0(  
  "https://raw.githubusercontent.com/CSSEGISandData/COVID-19/master/",  
  "csse_covid_19_data/csse_covid_19_time_series/time_series_covid19_",  
  "confirmed", # also: "deaths", "recovered"  
  "_global.csv"  
)
```

Read it in.

```
confirmed_global <- confirmed_global_url %>%  
  pins::pin() %>%  
  read_csv(col_types = cols(  
    .default = col_double(),  
    `Province/State` = col_character(),  
    `Country/Region` = col_character()  
  ))
```

province_or_state	country_or_region	Lat	Long	1/22/20	1/23/20	1/24/20	1/25/20	1/26/20	1/27/20	1/28/20
	Afghanistan	33.93911	67.709953	0	0	0	0	0	0	
	Albania	41.1533	20.1683	0	0	0	0	0	0	
	Algeria	28.0339	1.6596	0	0	0	0	0	0	
	Andorra	42.5063	1.5218	0	0	0	0	0	0	
	Angola	-11.2027	17.8739	0	0	0	0	0	0	

Notice that each day's count of confirmed cases is in a separate column. Suppose we want to plot the number of cases over time. What about the structure of this table might give us trouble?

province_or_state	country_or_region	Lat	Long	1/22/20	1/23/20	1/24/20	1/25/20	1/26/20	1/27/20	1/28/20
	Afghanistan	33.93911	67.709953	0	0	0	0	0	0	
	Albania	41.1533	20.1683	0	0	0	0	0	0	
	Algeria	28.0339	1.6596	0	0	0	0	0	0	
	Andorra	42.5063	1.5218	0	0	0	0	0	0	
	Angola	-11.2027	17.8739	0	0	0	0	0	0	

1-5 of 266 rows

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Notice that each day's count of confirmed cases is in a separate column. Suppose we want to plot the number of cases over time. What about the structure of this table might give us trouble?

How many observations does each row represent?

Think and discuss

Suppose we want to plot the number of cases over time.
What *should* the table look like?

Tidying Step 1: `pivot_longer`

- We have: Lots of observations per row
- We want: one observation per row.
- So: we're gonna need a *longer* (and narrower) dataset.

enter `pivot_longer`!

```
confirmed_global %>%  
  pivot_longer(  
    -(1:4) # the first 4 columns are not part of the pivot  
  )
```

```
## # A tibble: 66,234 x 6  
##   province_or_state country_or_region   Lat   Long name   value  
##   <chr>             <chr>         <dbl> <dbl> <chr>    <dbl>  
## 1 <NA>             Afghanistan    33.9   67.7 1/22/20     0  
## 2 <NA>             Afghanistan    33.9   67.7 1/23/20     0  
## 3 <NA>             Afghanistan    33.9   67.7 1/24/20     0  
## 4 <NA>             Afghanistan    33.9   67.7 1/25/20     0  
## 5 <NA>             Afghanistan    33.9   67.7 1/26/20     0  
## 6 <NA>             Afghanistan    33.9   67.7 1/27/20     0  
## 7 <NA>             Afghanistan    33.9   67.7 1/28/20     0  
## 8 <NA>             Afghanistan    33.9   67.7 1/29/20     0  
## 9 <NA>             Afghanistan    33.9   67.7 1/30/20     0  
## 10 <NA>            Afghanistan    33.9   67.7 1/31/20     0  
## # ... with 66,224 more rows
```

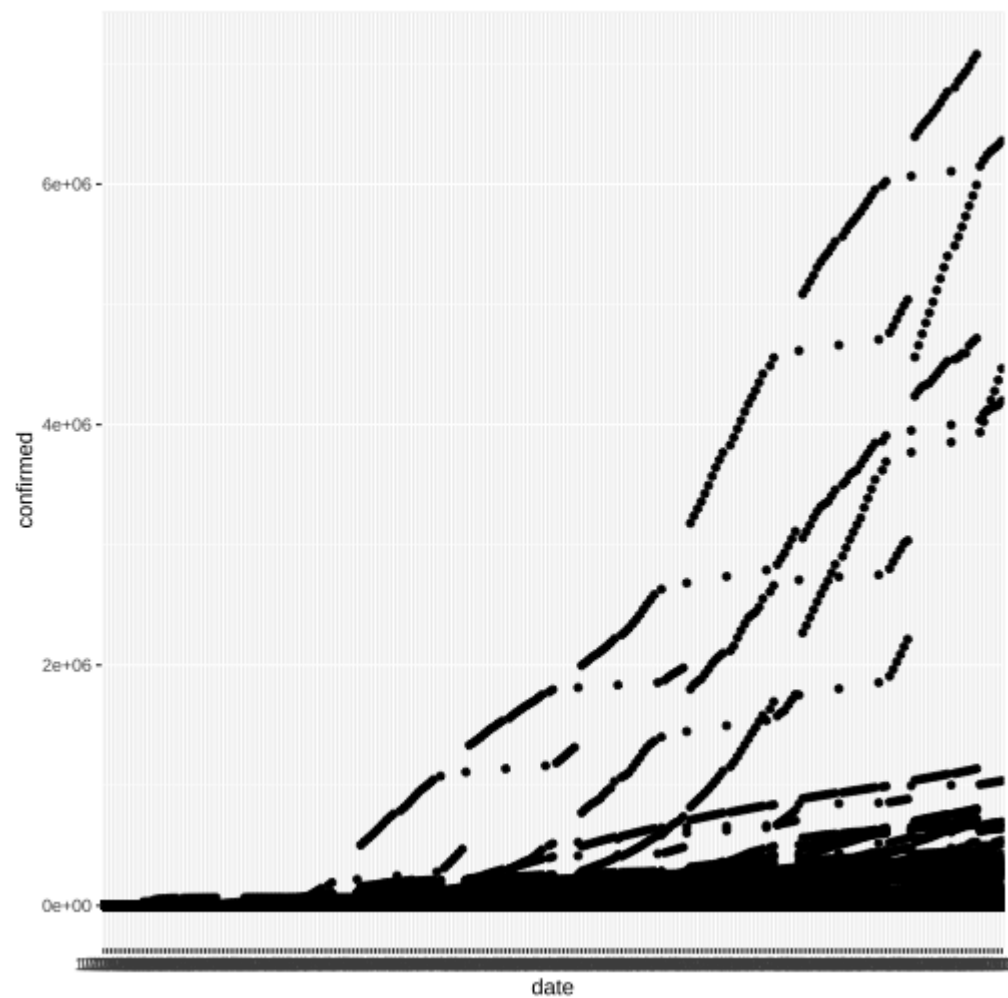
```
confirmed_global %>%
  pivot_longer(
    -(1:4),
    names_to = "date"
  )
```

```
## # A tibble: 66,234 x 6
##   province_or_state country_or_region   Lat   Long date      value
##   <chr>              <chr>          <dbl> <dbl> <chr>    <dbl>
## 1 <NA>              Afghanistan    33.9   67.7 1/22/20      0
## 2 <NA>              Afghanistan    33.9   67.7 1/23/20      0
## 3 <NA>              Afghanistan    33.9   67.7 1/24/20      0
## 4 <NA>              Afghanistan    33.9   67.7 1/25/20      0
## 5 <NA>              Afghanistan    33.9   67.7 1/26/20      0
## 6 <NA>              Afghanistan    33.9   67.7 1/27/20      0
## 7 <NA>              Afghanistan    33.9   67.7 1/28/20      0
## 8 <NA>              Afghanistan    33.9   67.7 1/29/20      0
## 9 <NA>              Afghanistan    33.9   67.7 1/30/20      0
## 10 <NA>             Afghanistan    33.9   67.7 1/31/20      0
## # ... with 66,224 more rows
```

```
confirmed_global_long <-
  confirmed_global %>%
  pivot_longer(
    -(1:4),
    names_to = "date",
    values_to = "confirmed"
  )
confirmed_global_long
```

```
## # A tibble: 66,234 x 6
##   province_or_state country_or_region  Lat  Long date      confirmed
##   <chr>              <chr>          <dbl> <dbl> <chr>         <dbl>
## 1 <NA>              Afghanistan    33.9  67.7 1/22/20         0
## 2 <NA>              Afghanistan    33.9  67.7 1/23/20         0
## 3 <NA>              Afghanistan    33.9  67.7 1/24/20         0
## 4 <NA>              Afghanistan    33.9  67.7 1/25/20         0
## 5 <NA>              Afghanistan    33.9  67.7 1/26/20         0
## 6 <NA>              Afghanistan    33.9  67.7 1/27/20         0
## 7 <NA>              Afghanistan    33.9  67.7 1/28/20         0
## 8 <NA>              Afghanistan    33.9  67.7 1/29/20         0
## 9 <NA>              Afghanistan    33.9  67.7 1/30/20         0
## 10 <NA>             Afghanistan    33.9  67.7 1/31/20         0
## # ... with 66,224 more rows
```

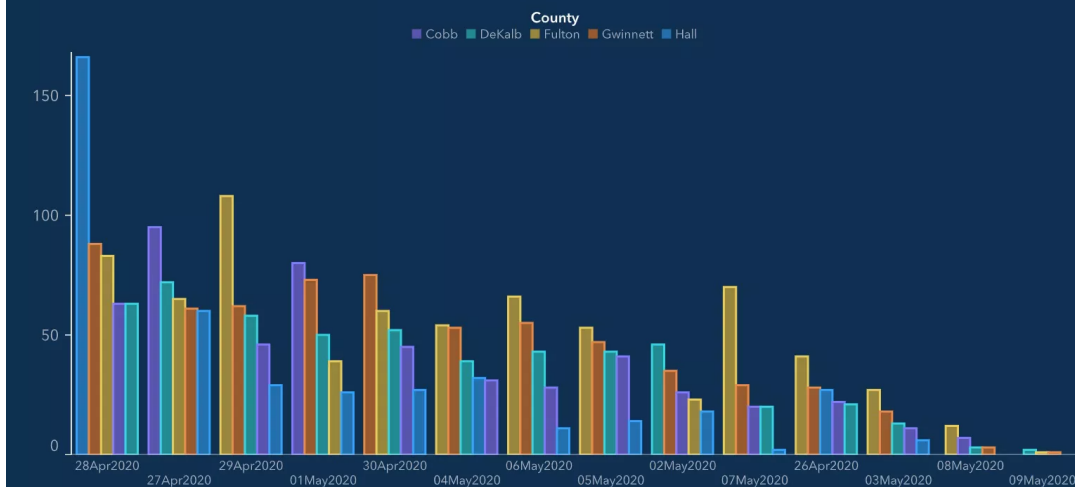
```
confirmed_global_long %>%  
  ggplot(aes(x = date, y = confirmed)) +  
  geom_point()
```



This happens!

Top 5 Counties with the Greatest Number of Confirmed COVID-19 Cases

The chart below represents the most impacted counties over the past 15 days and the number of cases over time. The table below also represents the number of deaths and hospitalizations in each of those impacted counties.



Source: <https://www.vox.com/covid-19-coronavirus-us-response-trump/2020/5/18/21262265/georgia-covid-19-cases-declining-reopening>

Aside: lots of Covid visualizations are problematic

- <https://sirota.substack.com/p/georgias-misleading-covid-map>
- <https://medium.com/nightingale/ten-considerations-before-you-create-another-chart-about-covid-19-27d3bd691be8>

```
"2020-02-01" %>%  
  parse_date() %>%  
  lubridate::month()
```

```
## [1] 2
```

```
"2/1/20" %>% parse_date() #<< Fail: date parser needs help!
```

```
"2/1/20" %>%  
  parse_date_time("%m/%d/%y!*") %>%  
  lubridate::month()
```

```
## [1] 2
```



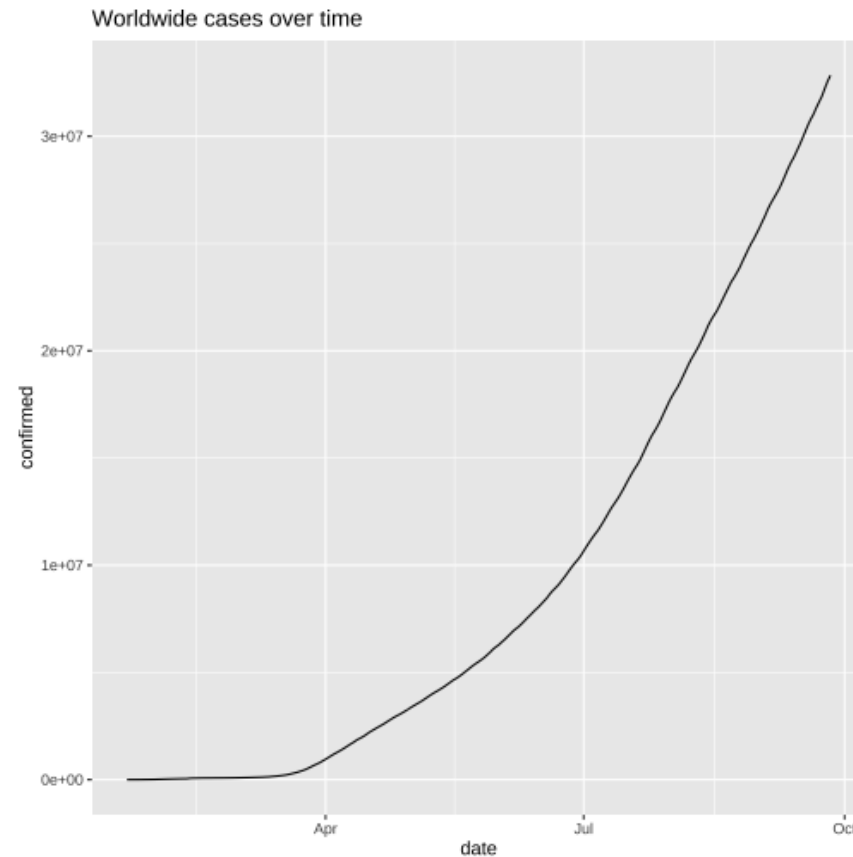
```
confirmed_global_long <-
  confirmed_global %>%
  pivot_longer(
    -(1:4), # the first 4 columns are not part of the pivot
    names_to = "date", # names of the remaining columns will be put into a date column
    values_to = "confirmed"
  ) %>% # values will be put into a column called confirmed
  mutate(date = lubridate::parse_date_time(date, "%m/%d/%y!*")) # convert to date objects

confirmed_global_long
```

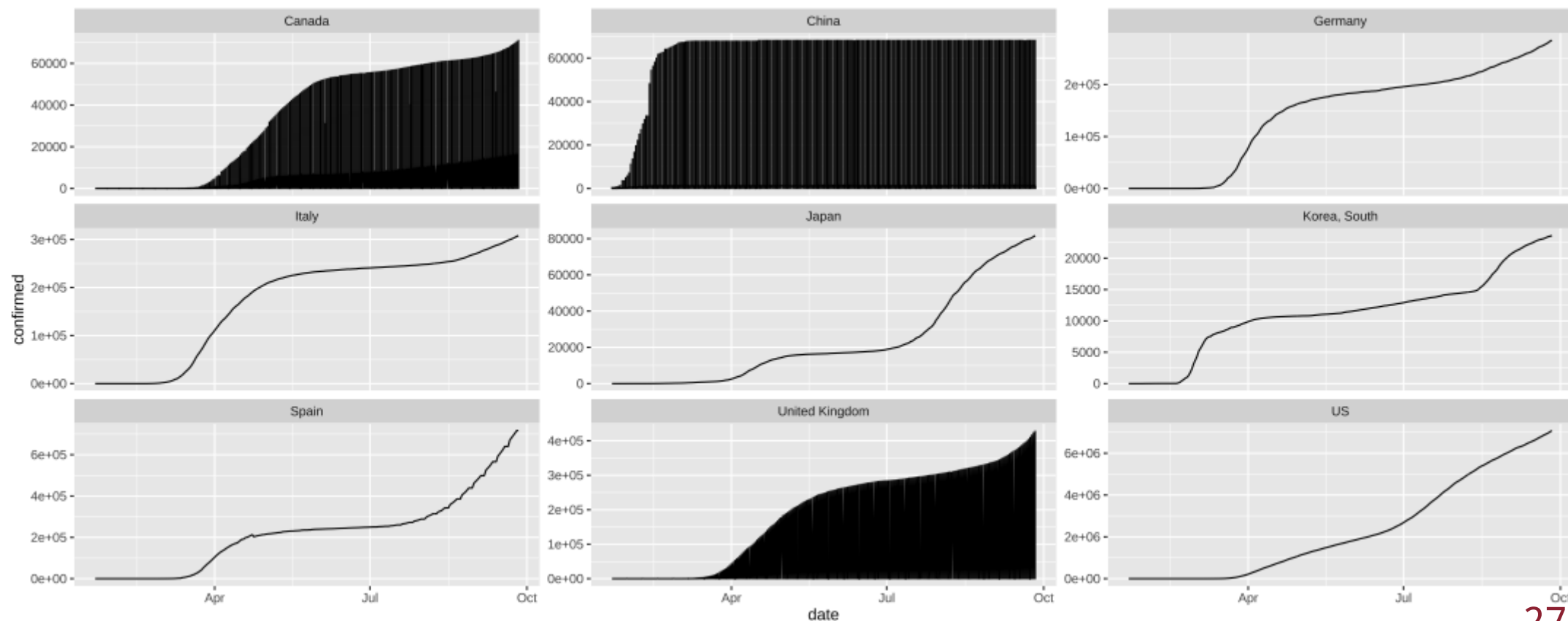
```
## # A tibble: 66,234 x 6
##   province_or_state country_or_region   Lat   Long date                confirmed
##   <chr>              <chr>         <dbl> <dbl> <dtm>                <dbl>
## 1 <NA>               Afghanistan  33.9  67.7 2020-01-22 00:00:00          0
## 2 <NA>               Afghanistan  33.9  67.7 2020-01-23 00:00:00          0
## 3 <NA>               Afghanistan  33.9  67.7 2020-01-24 00:00:00          0
## 4 <NA>               Afghanistan  33.9  67.7 2020-01-25 00:00:00          0
## 5 <NA>               Afghanistan  33.9  67.7 2020-01-26 00:00:00          0
## 6 <NA>               Afghanistan  33.9  67.7 2020-01-27 00:00:00          0
## 7 <NA>               Afghanistan  33.9  67.7 2020-01-28 00:00:00          0
## 8 <NA>               Afghanistan  33.9  67.7 2020-01-29 00:00:00          0
## 9 <NA>               Afghanistan  33.9  67.7 2020-01-30 00:00:00          0
## 10 <NA>              Afghanistan  33.9  67.7 2020-01-31 00:00:00          0
## # ... with 66,224 more rows
```

Plotting the data

```
confirmed_global_long %>%  
  group_by(date) %>%  
  summarize(confirmed = sum(confirmed)) %>%  
  ggplot(aes(x = date, y = confirmed)) +  
    geom_line() +  
    labs(title="Worldwide cases over time")
```



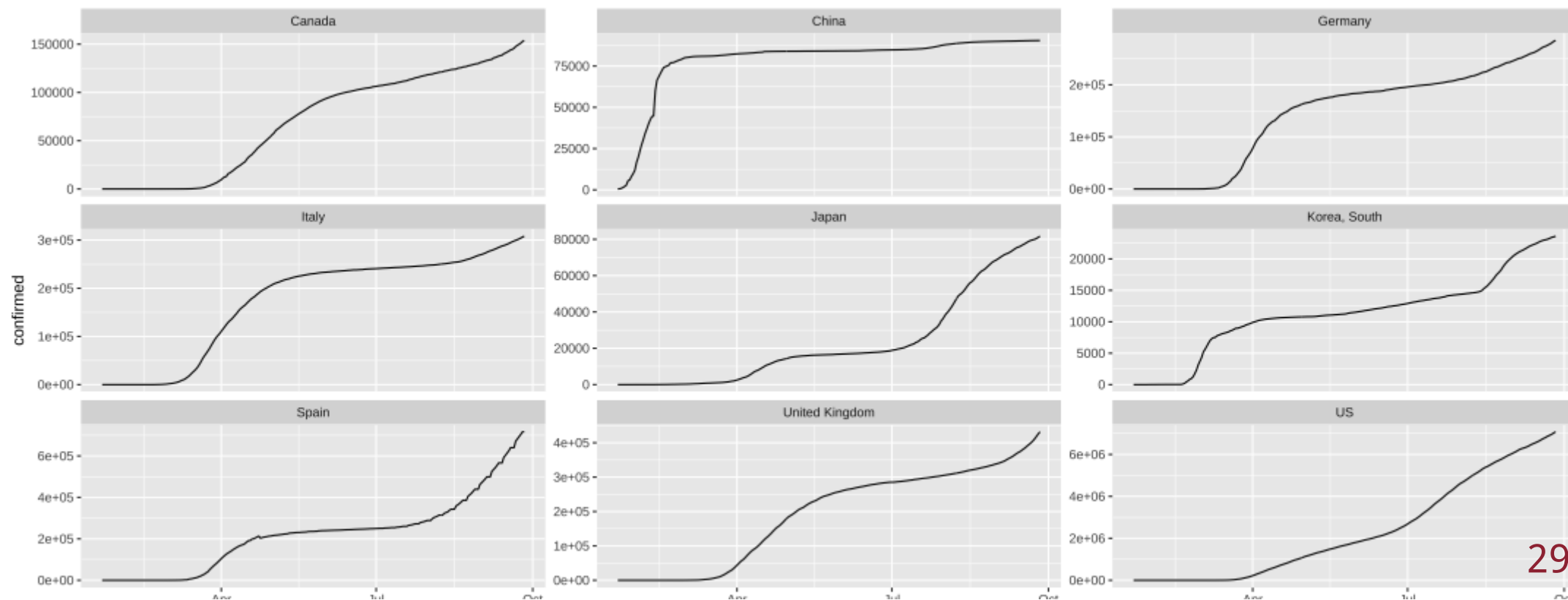
```
confirmed_global_long %>%
  filter(country_or_region %in%
    c("US", "Canada", "China", "Japan", "Korea, South", "Italy", "Germany", "Spain", "United Kingdom")
  )
ggplot(aes(x = date, y = confirmed)) +
  geom_line() +
  facet_wrap(~country_or_region, scales = "free_y")
```



```
confirmed_global %>%  
  count(country_or_region) %>%  
  filter(n > 1)
```

```
## # A tibble: 7 x 2  
##   country_or_region      n  
##   <chr>             <int>  
## 1 Australia          8  
## 2 Canada             14  
## 3 China              33  
## 4 Denmark            3  
## 5 France             11  
## 6 Netherlands         5  
## 7 United Kingdom     11
```

```
confirmed_global_long %>%
  filter(country_or_region %in%
    c("US", "Canada", "China", "Japan", "Korea, South", "Italy", "Germany", "Spain", "United Kingdom")
  )
  group_by(country_or_region, date) %>% # <<
  summarize(confirmed = sum(confirmed)) %>% # <<
  ggplot(aes(x = date, y = confirmed)) +
  geom_line() +
  facet_wrap(~country_or_region, scales = "free_y")
```



Per Capita?

A data source: the **World Bank**.

```
reload_data <- FALSE
if (reload_data) {
  wbstats::wb_data("SP.POP.TOTL",
    mrnev = 1 # most recent non-empty value
  ) %>% write_csv("data/worldbank_sp_pop_totl.csv")
}

population <-
  read_csv(
    "data/worldbank_sp_pop_totl.csv",
    col_types = cols_only(
      iso2c = col_character(),
      iso3c = col_character(),
      country = col_character(),
      date = col_double(),
      SP.POP.TOTL = col_double(),
      footnote = col_character()
    )
  ) %>%
  select(iso2c, iso3c, country, population = SP.POP.TOTL)
```

population %>% reactable::reactable()

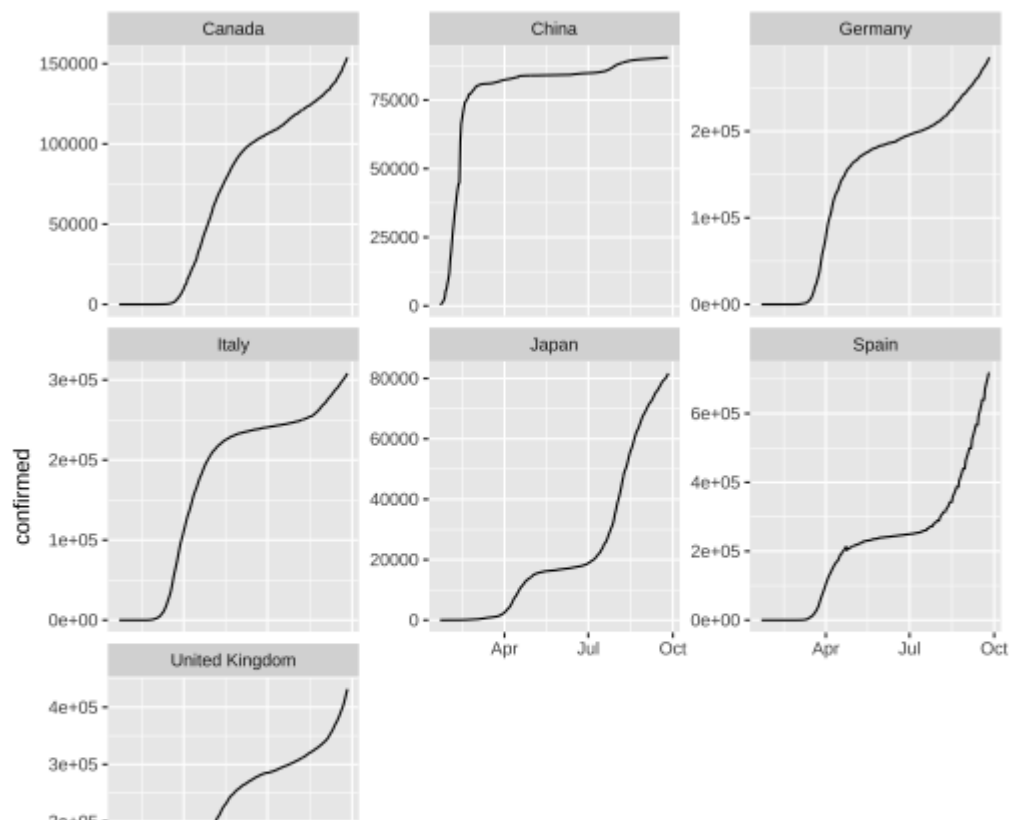
iso2c	iso3c	country	population
AW	ABW	Aruba	106314
AF	AFG	Afghanistan	38041754
AO	AGO	Angola	31825295
AL	ALB	Albania	2854191
AD	AND	Andorra	77142
AE	ARE	United Arab Emirates	9770529
AR	ARG	Argentina	44938712
AM	ARM	Armenia	2957731
AS	ASM	American Samoa	55312
AG	ATG	Antigua and Barbuda	97118

1–10 of 217 rows

Previous **1** 2 3 4 5 ... 22 Next


```
cases_with_population <- inner_join(  
  confirmed_global_long %>% select(country = country_or_region, date, confirmed),  
  population %>% select(country, population),  
  by = "country"  
)
```

```
cases_with_population %>%
  filter(country %in%
    c("US", "Canada", "China", "Japan", "Korea, South", "Italy", "Germany", "Spain", "United Kingdom")
  )
  group_by(country, date) %>% # <<
  summarize(confirmed = sum(confirmed)) %>% # <<
  ggplot(aes(x = date, y = confirmed)) +
  geom_line() +
  facet_wrap(~country, scales = "free_y")
```



Debugging a join

```
most_recent_cases <- confirmed_global_long %>%
  group_by(country = country_or_region) %>%
  slice_tail(n = 1)
full_join( # <<
  population %>% select(country, population),
  most_recent_cases %>% select(country, confirmed),
  by = "country"
) %>%
  filter(is.na(confirmed))
```

```
## # A tibble: 55 x 3
##   country      population confirmed
##   <chr>          <dbl>      <dbl>
## 1 Aruba          106314         NA
## 2 American Samoa    55312         NA
## 3 Bahamas, The     389482         NA
## 4 Bermuda           63918         NA
## 5 Brunei Darussalam 433285         NA
## 6 Channel Islands   172259         NA
## 7 Congo, Dem. Rep. 86790567        NA
## 8 Congo, Rep.       5380508         NA
## 9 Curacao          157538         NA
## 10 Cayman Islands    64948         NA
## # ... with 45 more rows
```

Recoding

```
recoded_cases <-  
  confirmed_global_long %>%  
  mutate(country = case_when(  
    country_or_region == "Russia" ~ "Russian Federation",  
    country_or_region == "US" ~ "United States",  
    country_or_region == "Korea, South" ~ "Korea, Rep.",  
    TRUE ~ country_or_region  
  ))  
cases_with_population <- inner_join(  
  recoded_cases %>% select(country, date, confirmed),  
  population %>% select(country, population),  
  by = "country"  
)
```

```
cases_with_population %>%
  filter(country %in%
    c("United States", "Canada", "China", "Japan", "Korea, Rep.", "Italy", "Germany", "Spain", "United Kingdom"))
  group_by(country, population, date) %>% # <<
  summarize(confirmed = sum(confirmed)) %>% # <<
  ggplot(aes(x = date, y = confirmed / population)) +
  geom_line() +
  facet_wrap(~country, scales = "fixed")
```

